

# Closed-Loop Dynamic Thermal Management (DTM) for 3D Stacked HBM2 Architectures

A Comprehensive Engineering and Implementation Journey

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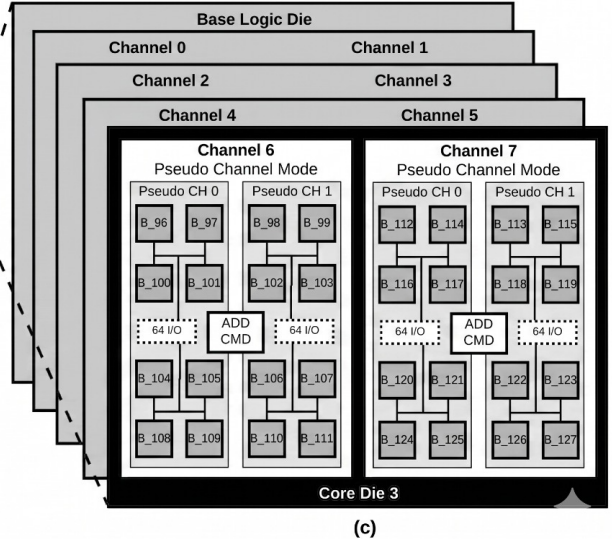
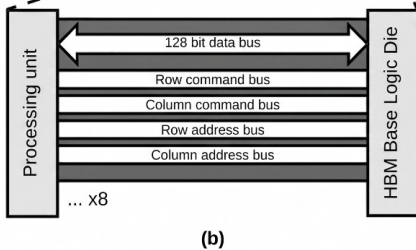
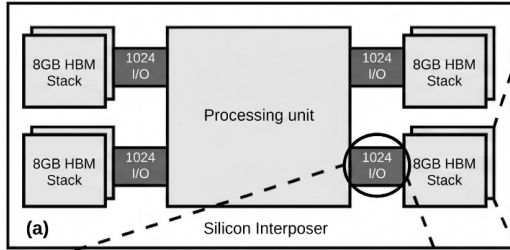
## The Challenge with HBM2:

- Transition from 2D (GDDR6) to 3D stacked High Bandwidth Memory (HBM) revolutionized GPU bandwidth.
- **Thermal Constraints:** Stacking active silicon dies vertically introduces severe heating.
- High temperatures (e.g., 85°C) destroy data retention in memory capacitors.

## Project Goal:

- Simulate thermal dissipation and real-time throttling of a HBM2 stack under intense workloads.
- Create an accurate Dynamic Thermal Management (DTM) policy scheduler based on generated power traces.

# HBM memory architecture



# The Simulation Pipeline

The project was broken into four distinct phases to handle the multi-domain complexity:

- 1 **Microarchitectural Simulation:** Generate realistic DRAM access patterns.
- 2 **Power Modeling:** Translate raw DRAM commands into physical wattage.
- 3 **3D Thermal Modeling:** Build a physical HBM stack representation to calculate spatial heat flow.
- 4 **Closed-Loop Control:** Architect a software DTM controller to monitor and recursively throttle the simulation.

# Phase I: Microarchitectural Trace Generation

## Tooling & Setup:

- Used Accel-Sim / GPGPU-Sim with NVIDIA's Volta V100 (QV100-SASS) architecture.
- **Workload:** SRAD (Speckle Reducing Anisotropic Diffusion) from Rodinia 3.1.
- **Run specifics:** `sim_run_12.8/srad_v1-rodinia-3.1/100_0_5_502_458/QV100-SASS`
- Chosen due to heavy spatial memory access patterns, yielding a lot of activity to record and triggering DTM thresholds.

## Rectifying HBM2 vs GDDR6:

- Expanded bus width to 16 bytes per channel.
- Reduced Burst Length (BL) to 2.
- Locked core clock to 850 MHz (256 bits per command).
- 8 partitions per stack, 16 banks per partition → 128 memory banks.

## Phase II: Bridging Architecture to Power

### The Power Formula:

$$P_{bank} = \frac{(C_{act}E_{act}) + (C_{pre}E_{pre}) + (C_{rd} \cdot B \cdot E_{rd}) + (C_{wr} \cdot B \cdot E_{wr})}{\Delta T}$$

$C$  = Command count,  $B = 256$  bits/command,  $\Delta T \approx 1.176$  ns/cycle

# Phase III: 3D Thermal Modeling (HotSpot 3D)

## Layer Configuration (LCF) for Stack:

- 9 total vertical layers configured.
- **4 Active Silicon Dies** (power dissipation enabled).
- **5 Inactive TIM/Glue Layers** (power dissipation disabled).

## Data Integration Quirks:

- TIM layers were completely ignored by HotSpot's parser. Python generator adjusted to output exactly 128 columns (32 banks  $\times$  4 layers).
- **Naming Convention:** Used unique bank names (B\_0 to B\_127) to bypass HotSpot's layer prefix duplication logic.

# Phase IV: Data Engineering & Optimization

## The Bottleneck:

- Seconds of logical execution inflated CSV trace data to **30 GB - 100 GB**.
- Standard Python loops failed under RAM constraints (16GB System RAM, 4GB VRAM).

## Out-of-Core Streaming with Polars:

- Replaced standard `for`-loops with **Polars** (Rust-based DataFrame library).
- Used LazyFrames and streaming execution to parse data *out-of-core* directly from SSD.
- Processed 30 GB CSVs into 8 independent partition files without loading into system RAM.
- **Result:** Processing time plummeted from hours to **under 3 minutes**.



# Phase V: Closed-Loop DTM Controller

## The DTM Policy:

- **Throttle ON:** Breaching 355.00 K (82°C).
- **Throttle OFF:** Cooling below 353.00 K (80°C).

## The $O(N^2)$ Trap vs. $O(1)$ “Stepper Motor”:

- Growing trace files recursively forced HotSpot into  $O(N^2)$  recalculations.
- **Solution:** Developed a stateful “Stepper Motor”. Passes exactly **1 row** of power data with an `.init` file representing previous thermal state.
- Loop execution time slashed to fractions of a second.

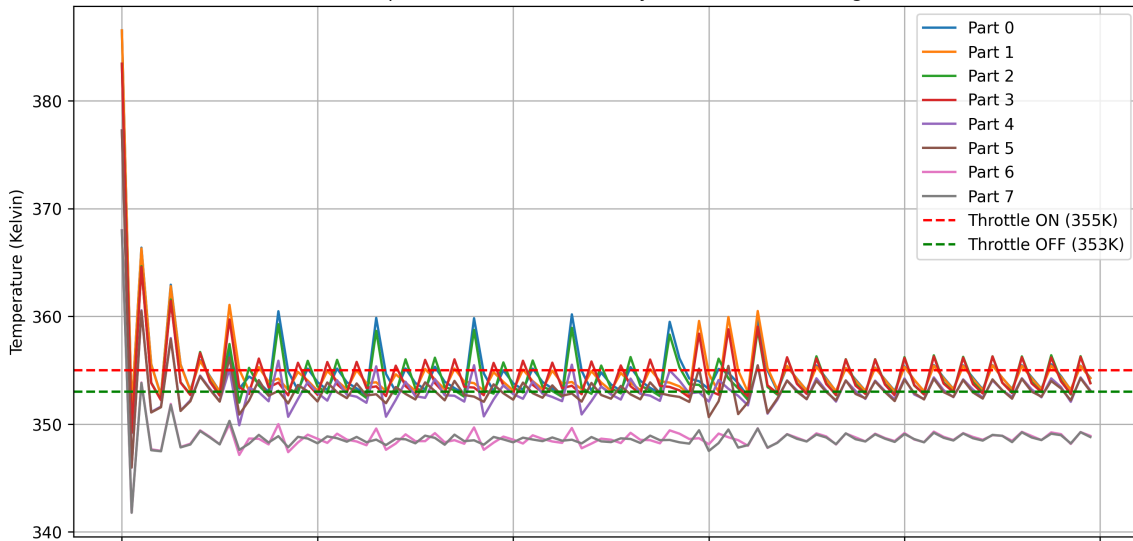
# Phase V: Asynchronous Throttling Execution

## Asynchronous Partition Pointers:

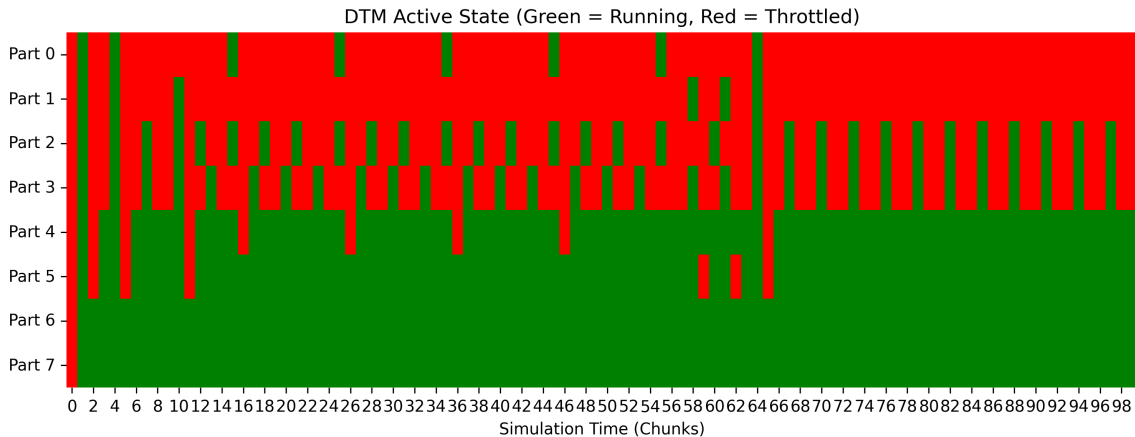
- Handled throttling without desynchronizing memory trace data.
- Maintained 8 independent file handles with an `active_state` array.
- If Partition  $P_i$  is throttled, inject 0.0W array for  $P_i$  and leave its file pointer stationary.
- Perfectly mimics a stalled memory controller clock.

# DTM Results

HBM2 Temperature over Time with Dynamic Thermal Management



# Actual trace file



# Conclusion and Outcomes

- Built a robust, scalable, software-defined 3D thermal simulation framework for HBM2 architectures.
- Successfully generated high-activity traces from the SRAD benchmark on QV100-SASS.
- Solved massive data bottlenecks using a multithreaded **Polars** pre-aggregator.
- Engineered an optimized  $O(1)$  Python stepper loop to interface with HotSpot 3D physics.
- The project lays a critical foundation for analyzing physical performance degradation and designing thermal-aware memory controllers.

# Thank You

Questions?